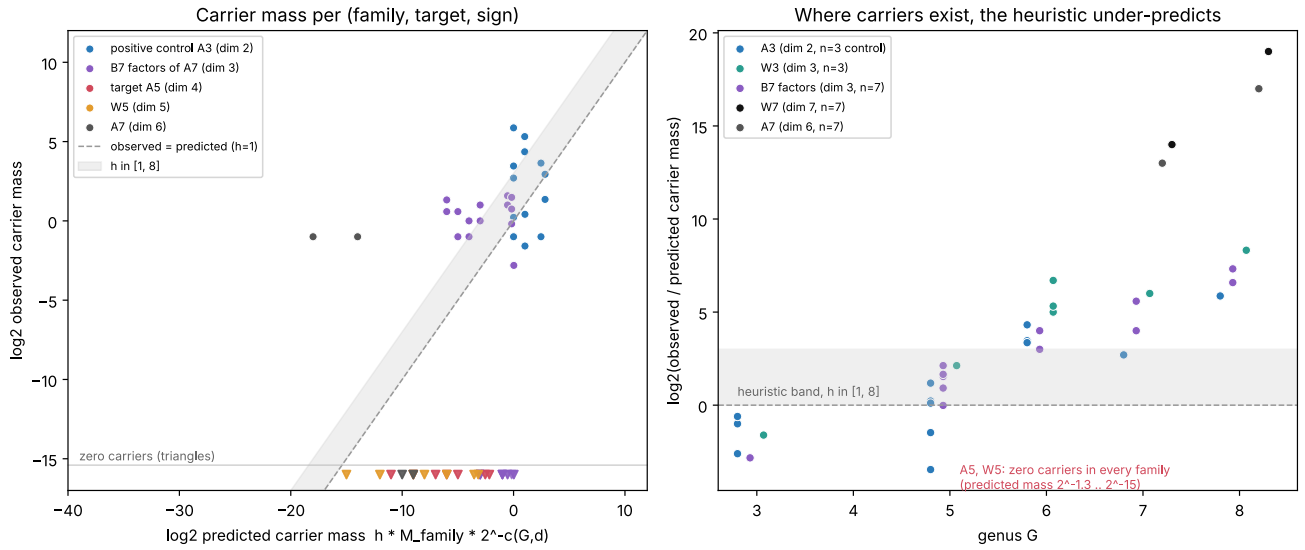
Torelli toy ladder (item 10)

Item 10: the Torelli rarity heuristic on the n = 5 and n = 7 toy ladder

Verdict: NEGATIVE at n = 5, but with low power. At n = 7 the literal reopen criterion is met, by a mechanism that n = 131 lacks.

- **n = 5.** I enumerated every genus-5 curve over F_2: hyperelliptic and trigonal completely, and canonical curves by a uniform sample of 2^{21} nets. No Jacobian among them carries `A_5`, for either sign. The same holds for every smooth plane quintic and every hyperelliptic curve of genus 6, 7 and 8. By the report's kill criterion this "confirms rarity". However, the heuristic itself predicts only about 0.42 classes per sign at genus 5 (h = 1), so a zero is what it expects. This zero tests little.
- **n = 7.** Hyperelliptic curves of genus 7 = dim A + 1 exist whose Jacobian is isogenous to the full Weil restriction `W_7 = Res_{F_128/F_2}(E_a)`, and so carries `A_7`. There are 1,536 models (mass 0.5) for each sign, and more at genus 8. The heuristic predicted 2^{-14} to 2^{-18} for these. This meets the literal criterion "more than one carrier at genus ≤ dim A + 2 reopens the route".
  - n = 7 is not a faithful analogue of 131. $\mathrm{ord}_7(2) = 3$, so `t^7 - 1` has cubic factors over F_2, and `A_7` is not even simple: it splits into two 3-dimensional factors.
  - The genus $7 = 2^3 - 1$ equals what GHS descent gives with magic number 4. That mechanism needs small factors of `t^n - 1`, and at n = 131 these do not exist. Attributing the genus-7 carriers to GHS is an interpretation; I did not construct the descent.
- **The heuristic as a quantitative predictor.** For the `A_3` control it is right to within $2^{-3.5}$ to 2^{+1.2} at G ≤ 5, which is excess genus G − d ≤ 3. At larger G it under-predicts every structured (Weil-restriction-type) class that occurs: by 2^{2.7} to 2^{8.3} at n = 3 (G = 6–8) and by up to 2^{19} at n = 7. Its codimension penalty for the extra genus is not what happens for these classes.
- **What this means for ECC2K-130.** These toys cannot confirm the heuristic. At n = 5, the one faithful rung, it predicts nothing and nothing is seen. Wherever carriers do exist, they come from structure that the dimension count ignores. The ECC2K-130 rarity argument therefore rests on the absence of such structure at n = 131 (no small magic numbers, simple A), not on the codimension number 8128.



carrier mass vs prediction

Left: observed against predicted carrier mass for every (family, target, sign). Triangles at the bottom are zero observations. Right: $\log_2$(observed/predicted) for every nonzero observation. This panel is selected on nonzero counts, so it shows where carriers exist, not an unbiased calibration.

What was tested

- Targets. `A_n` is the trace-zero part of `Res_{F_{2^n}/F_2}(E_a)` for `E_a : y² + xy = x³ + ax² + 1`, of dimension n − 1, with characteristic polynomial `P_{A_n} = W_n / (T² - tT + 2)`, where `W_n = T^{2n} - s_n T^n + 2^n`.
- Detection. A curve "carries" a target when the target polynomial divides `P_C`. `P_C` comes from point counts over F_{2^k}, k ≤ g.
- PARI/GP structure checks:
  - `P_{A_3}` and `P_{A_5}` are irreducible.
  - `P_{A_7} = B_7 · B'_7`, two degree-6 factors, for each sign.
  - All CM class numbers are $h_K = h_K^+ = 1$.
  - $[O_K : \mathbb{Z}[\pi]] = 2$, 64 and 8 for n = 3, 5 and 7.
- Heuristic (frozen in PROTOCOL.md): the predicted carrier mass is `pred = h · M_family · 2^{-c(G,d)}`, with `c = dG - d(d-1)/2`. This is the codimension in A_G of the locus {A × B}. At G = d = 130 it gives the note's 8128.

- Mass is the number of models divided by the order of the group acting on models. This equals Σ 1/|Aut| over isomorphism classes.

Families and coverage

arm	genus	family	models	smooth	group	mass M
hyp3	3	hyperelliptic, all	15,872	6,144	192	32
quartic	3	plane quartics, all	32,768	10,920	168	65
hyp5	5	hyperelliptic, all	1,040,384	393,216	768	512
trigonal5	5	plane quintics with one δ=1 double point at [0:0:1], all	262,144	67,608	24	2,817
ci5	5	canonical: smooth intersections of 3 quadrics in P^4, 2^{21} uniform nets (stage-1); 2^{13} subsample fully counted	2,097,152	1,422 / 8,192 in subsample (17.36% ± 0.42%)	9,999,360	3,635 ± 88
hyp6	6	hyperelliptic, all	8,355,840	3,145,728	1,536	2,048
quintic	6	smooth plane quintics, all	2,097,152	688,128	168	4,096
hyp7	7	hyperelliptic, all	66,977,792	25,165,824	3,072	8,192
hyp8	8	hyperelliptic, all	536,346,624	201,326,592	6,144	32,768

- **Genus 5 is complete up to the canonical sample.** Hyperelliptic, trigonal and canonical curves partition M_5, and the total mass is 6,964 ≈ 2^12.8.
    - Sampling cost: each canonical net carries mass 0.0100. A class with |Aut| = a is expected to be hit about 100/a times, so only classes with |Aut| ≳ 30 can be missed with noticeable probability.
- **Genus 6 is partial.** It covers mass 6,144 against dim M_6 = 15, i.e. ≈ 2^{15} curves. Missing are trigonal genus 6, the general genus-6 curve on a del Pezzo surface, and the bielliptic locus.
- **Genus 7 and 8 are hyperelliptic only.**

Hyperelliptic models are $y^2 + hy = f$ with deg h ≤ g + 1 and deg f ≤ 2g + 2. They are reduced modulo y ↦ y + e(x), so each representative stands for 2^{g+1} models.

Results

Each entry is the number of carrier models, then mass and predicted mass (h = 1) per sign. “a / b” gives E_0 / E_1. “—” means the target dimension exceeds the genus.

arm	G	A3 (control)	W3	B7, B’7 (dim 3)	A5 (dim 4)	W5 (dim 5)	A7 (dim 6)	W7 (dim 7)
hyp3	3	96 / 96 (0.5 each; pred 1)	0	0	—	—	—	—
quartic	3	56 / 224 (0.33 / 1.33; pred 2.0)	0 / 56 = Klein	E_1 only: 24, 24 (pred 1.0)	—	—	—	—
hyp5	5	896 / 896 (1.17; pred 1)	0	0	**0** (pred 0.031)	0 (0.016)	—	—
trigonal5	5	12 / 300 (0.5 / 12.5; pred 5.5)	0 / 72	72 (B’7 E_0), 48 (B’7 E_1)	**0** (pred 0.17)	0 (0.086)	—	—
ci5	5	1 / 3 of 1,422 sampled (pred 7.1)	sample 0 / 1	11–35 hits on 2^{18} nets	**0 on 2^{21} nets** (pred 0.22; 95% upper 0.030)	0 (0.11)	—	—
hyp6	6	16,896 each (11.0; pred 1)	3,072 each	768–1,536	**0** (0.0078)	0	0 (0.00098)	—
quintic	6	3,444 / 6,692 (20.5 / 39.8; pred 2)	840 / 2,184	168–336	**0** (0.016)	0	0 (0.0020)	—
hyp7	7	19,968 each (6.5; pred 1)	6,144 each	1,536–4,608	**0** (0.0020)	0	**1,536 each (0.5; pred 6.1e-5)**	**1,536 each (0.5; pred 3.1e-5)**

arm	G	A3 (control)	W3	B7, B'7 (dim 3)	A5 (dim 4)	W5 (dim 5)	A7 (dim 6)	W7 (dim 7)
hyp8	8	358,400 each (58.3; pred 1)	30,720 each	9,216–15,360	**0** (0.00049)	0	**3,072 each (0.5; pred 3.8e-6)**	3,072 each (0.5; pred 9.5e-7)

Full rows, with family masses, predictions and Poisson zero-probabilities, are in `results_table.csv` and `results.json`.

n = 5: the faithful rung

- **Genus 5.** No A_5 or W_5 carrier exists among hyperelliptic or trigonal genus-5 curves (both complete). None occurs among 2,097,152 sampled canonical nets.
  - 91,740 nets passed the (N_1, N_2, N_3) filter and 14,210 the N_4 filter. All were checked fully, with 0 hits.
  - For canonical curves the carrier mass is < 0.030 at 95% (zero events in 2^{21} nets, at 0.0100 mass per net).
- **Genus 6.** None among smooth plane quintics (complete) or hyperelliptic genus-6 curves (complete), and none at genus 7 or 8 among hyperelliptic curves.
- **Predicted counts.**
  - The heuristic predicts a genus-5 carrier mass of 0.42 per sign at h = 1, so 0.85 for both signs. P(0) is 0.43 at h = 1 and 0.001 at h = 8.
  - The ppav count h of A_5's class is not computed. [O_K : Z[π]] = 64 allows several orders, so h could exceed 1. If h ≥ 4, the genus-5 zero is significant at about 3%. That condition is not established.
  - At genus 6, over the covered mass, the prediction is 0.023 per sign, so the zero there is uninformative.
- **Comparison with occurring factors.** In the genus-5 families, 135 (hyp5), 276 (trigonal) and 175 (canonical subsample) distinct simple 4-dimensional factor classes occur, with median mass 0.5–2.6. A_5 is not among them. I did not count the total number of 4-dimensional isogeny classes over F_2, so no rank or percentile is claimed.

n = 7: unfaithful rung, literal reopen

- **The curves.** Genus-7 hyperelliptic curves with $P_C = W_7 = T^{14} \mp 13T^7 + 128$ exist for both signs: 6 reduced representatives each, so 1,536 models and mass 0.5.
  - Example for E_0: $h = x^7 + x^5 + x^4 + x^3 + 1$, $f = x^{15} + x^6 + x^4 + 1$.
- **Independent check** (`verify_hits.py`, `n7_carriers_check.json`). A brute-force count of $y^2 + hy = f$ over F_{2^8}, F_{2^9} and F_{2^10} gives 257, 513 and 1025 points for both examples. This equals the prediction from W_7, beyond the k ≤ 7 counts used to build P_C.
- **What A_7 is.** It is $B_7 \times B'_7$, and each 3-dimensional factor already occurs at genus 3 (the E_1 factors in plane quartics) and genus 5. With A not simple and $\mathrm{ord}_7(2) = 3$, this rung tests a different situation from n = 131.

The positive control and the heuristic's calibration

- **Instrument control.** The instrument reproduces the 2026-10-06 ladder exactly:
  - smooth-model counts 768, 6,144 and 393,216;
  - A_3 hits 96/96 (hyp3), 56/224 (quartic) and 896/896 (hyp5);
  - the Klein quartic, as $W_3(E_1)$, with 56 models;
  - 10,920 smooth quartics, i.e. mass 65 = $q^6 + 1$;
  - the hyperelliptic masses q^{2g−1}.
- **Calibration at G ≤ 5.** Observed/predicted for A_3 lies between $2^{-3.5}$ (trigonal, E_0: 0.5 against 5.5) and 2^{+1.2} (trigonal, E_1: 12.5 against 5.5). So the heuristic is right to within its own h-band and a factor of a few.
- **Under-prediction at G = 6–8.** The ratio is 2^{2.7} to 2^{5.9} for A_3 and 2^{5} to 2^{8.3} for W_3. The prediction falls as 2^{−dG} relative to the family mass, while the observed carrier mass stays flat or grows.
- Every L-polynomial in the complete arms is a valid Weil polynomial: the place counts for k ≤ 2g are non-negative integers.

Deviations from the protocol

1. **Canonical genus 5 was sampled, not enumerated modulo PGL_5.**
   - The design asked for exhaustive orbit enumeration. I sampled 2^{21} uniform nets instead (mass 0.0100 per net, coverage as above).
   - The B_7 filter ran on the first 2^{18} nets only, and the full-L-polynomial subsample is 2^{13} nets.
   - So the A_3 control in the canonical arm rests on 4 hits.
2. **Smoothness is a search for singular points over F_{2^k}, not a full check.**
   - k ≤ 6 for plane curves. This is rigorous for irreducible curves of degree ≤ 5. For reducible curves it relies on intersection orbits of size ≤ 6.
   - k ≤ 5 for canonical nets. Of the 1,422 subsample nets accepted as smooth, 9 (0.6%) produced invalid Weil polynomials, so they were singular or not curves. They are excluded from the target counts and inflate the canonical mass by 0.6%.
   - The hyperelliptic gcd criterion is exact.
   - None of this can hide a carrier: a curve wrongly rejected as singular would need a real singular point.
3. **Genus 6–8 are not complete.** Genus 6 covers plane quintics and hyperelliptic curves only; genus 7 and 8 are hyperelliptic only. The n = 7 target "through genus 8" is met for the hyperelliptic locus only. The GHS magic-number-4 cover at n = 5 (genus 7 or 8, expected non-hyperelliptic) lies outside the searched families.

4. **h_A was not computed.** It is reported as a band, h ∈ [1, 8].
5. **The run was slower than planned.** The machine was shared by ten experiments (load 11–29 on 4 cores). The first canonical run was restarted with an N_4 pre-filter. No numbers were lost.

Reproduce

```
        cd research/ic_next10_20261014/10-torelli-toy
        OMP_NUM_THREADS=1 nice prlimit --as=700000000 -- python3 -I run_arms.py hyp2 hyp3 quartic hyp5 trigonal5 hyp6 hyp7 hyp8
quintic   # ~7 min
        OMP_NUM_THREADS=1 nice prlimit --as=700000000 -- python3 -I run_ci.py 21 13 data/g5_ci.json
# ~22 min (seed 20261014)
        OMP_NUM_THREADS=1 nice prlimit --as=700000000 -- python3 -I analyze.py      # results.json, results_table.csv (needs
gp)
        python3 -I verify_hits.py                                                    # n7_carriers_check.json
        python3 -I plot.py                                                           # torelli_toy.svg / .png
```

Code: - `gf2k.py`: binary fields. - `lpoly.py`: L-polynomials and targets. - `fam_hyper.py`, `fam_plane.py`, `fam_ci.py`: the families.

Raw histograms of (N_1..N_g) per arm are in `data/` (4 MB).